MALL CUSTOMER SEGMENTATION & SPENDING BEHAVIOUR ANALYSIS

A Comprehensive Business Intelligence Report on
Customer Demographics, Income Distribution & Spending Patterns

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Dataset: Mall Customers | Records: 200 | Classification: Confidential

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Executive Summary

This report presents a thorough business intelligence analysis of a retail mall's customer base, examining 200 shoppers across dimensions of gender, age, annual income, and spending behaviour. The primary objective is to identify actionable customer segments that can inform targeted marketing strategies, loyalty programme design, and revenue optimisation initiatives.

200	56.3%	62.3k	38.6
Total Customers	Female Ratio	Avg. Annual Income ($k)	Average Age
13.9	**~50**	**32**	**5**
Std Dev of Age	Median Spending Score	Golden Customers	Behavioural Clusters

Key Findings at a Glance

- The customer base is majority female (56.3%), a critical factor for product assortment and campaign design.
- The 25–35 age cohort represents the highest footfall, making them the primary acquisition and retention target.
- Annual income peaks at the $55–70k band, which correlates strongly with high spending scores.
- 32 'Golden Customers' — aged 23–55, income above $55k, spending score ≥ 75 — represent the highest-value segment.
- A measurable low-value fringe (spending score ≤ 25) exists, signalling potential churn or disengagement.
- Mid-range spenders (score 36–69) form the largest cohort, offering the greatest upselling opportunity.

SECTION 2

Dataset Overview & Methodology

The analysis is based on the publicly available Mall Customers dataset, which captures key socio-demographic and behavioural attributes for 200 unique customers. All analysis was performed in Python using pandas and NumPy, with visualisations produced in Power BI.

Dataset Schema

Field	Type	Range / Values	Description
CustomerID	Integer	1 – 200	Unique customer identifier
Gender	Nominal	Male / Female	Customer gender
Age	Integer	18 – 70	Customer age in years
Annual Income (k$)	Integer	15 – 137	Annual income in USD thousands
Spending Score (1-100)	Integer	1 – 99	Mall-assigned spending behaviour score

Analytical Approach

The analysis proceeded through three phases: (1) descriptive statistics and distribution profiling, (2) rule-based segmentation using domain-defined thresholds on income and spending score, and (3) dashboard visualisation in Power BI to enable interactive exploration by business stakeholders.

> ■ **Insight:** Spending Score is a composite index assigned by the mall based on purchase frequency, basket size, and loyalty programme engagement — making it a reliable proxy for customer value.

SECTION 3

Customer Demographics Analysis

3.1 Gender Distribution

Of the 200 customers surveyed, 112 (56%) are female and 88 (44%) are male. This female majority is significant for visual merchandising, product placement, and communication strategy — female-oriented campaigns are likely to yield higher ROI.

Segment	Count	Proportion	Avg Spending Score
Female	112	56.00%	51.5
Male	88	44.00%	48.5
Total	200	100.00%	50.2 (overall avg)

■ **Insight:** Female customers slightly outperform males on average spending score, reinforcing the case for female-first campaign prioritisation.

3.2 Age Distribution

The average customer age is 38.6 years with a standard deviation of 13.9, indicating a broad demographic spread. The modal age group (25–34) represents over 35% of footfall, followed by the 35–44 cohort. Customers under 25 and over 60 account for relatively small shares.

Age Band	Approx. Count	% of Total	Marketing Priority
18 – 24	25	12.5%	Growth Segment
25 – 34	72	36.0%	Core / Primary
35 – 44	51	25.5%	High Value
45 – 54	30	15.0%	Retention Focus
55 – 70	22	11.0%	Secondary

■ **Insight:** The 25–44 age band collectively accounts for 61.5% of all customers. Retention and upselling efforts concentrated here will have maximum portfolio impact.

SECTION 4

Spending Score Distribution

The Spending Score (1–100) is the central behavioural metric in this dataset. It encapsulates a customer's purchasing intensity and is directly correlated with revenue contribution. Analysis reveals a trimodal pattern with concentrations in the low (≤35), mid (36–69), and high (≥70) bands.

4.1 Score Band Summary — By Gender

Score Band	Male Count	Female Count	Total	Share of Total
Low (≤ 35)	40	38	78	39.0%
Mid (36–69)	38	59	97	48.5%
High (≥ 70)	10	15	25	12.5%
Total	88	112	200	100%

■ **Insight:** Nearly half of all customers occupy the mid-range band — a prime target for score elevation through loyalty incentives, personalised promotions, and exclusive events.

4.2 Income vs. Spending Score Relationship

The scatter distribution from the Power BI dashboard reveals a non-linear relationship between income and spending score. Several distinct clusters emerge: high-income / high-spenders (premium), high-income / low-spenders (frugal affluent), low-income / high-spenders (impulsive), and low-income / low-spenders (budget-conscious). This bifurcation at mid-income levels (around $50–70k) reflects divergent spending philosophies.

SECTION 5

Customer Segmentation & Clusters

Rule-based segmentation was applied using income and spending score thresholds to classify customers into three actionable tiers. This framework enables differential marketing investment and service-level decisions.

5.1 Segment Definitions

Segment	Criteria	Size	Female	Male	Priority
Golden (High Value)	Age 23–55 | Income >$55k Spend Score ≥ 75	32	~19	~13	Retain & Delight
Core (Mid Value)	Score 36–69 (All income levels)	97	~59	~38	Upsell & Engage
At-Risk (Low Value)	Score ≤ 25 | Age <23 or >55 Income >$50k	Varies	—	—	Re-engage / Win-back

5.2 Golden Customer Profile (Deep Dive)

The 32 Golden Customers represent the highest-value cohort: working professionals aged 23–55 with disposable incomes above $55,000 and a demonstrated tendency to spend. Female customers represent approximately 59% of this elite group, consistent with the broader gender distribution but slightly more pronounced at higher engagement levels.

- Average Spending Score: ~83 (significantly above population mean of 50)
- Average Annual Income: ~$80k+
- Female share within segment: ~59%
- Primary age concentration: 28–42 years
- Recommended strategy: VIP programme, early access, concierge services

■ **Insight:** Even a 10% increase in Golden Customer visit frequency could generate a disproportionate lift in overall mall revenue. Priority investment here yields the highest ROI.

SECTION 6

Dashboard Visualization

The interactive Power BI dashboard below provides a consolidated view of the customer data, featuring dynamic slicers for spending score range and gender. Key components include a demographic age histogram, a gender pie chart, KPI cards for aggregate metrics, and an income vs. spending score distribution chart.

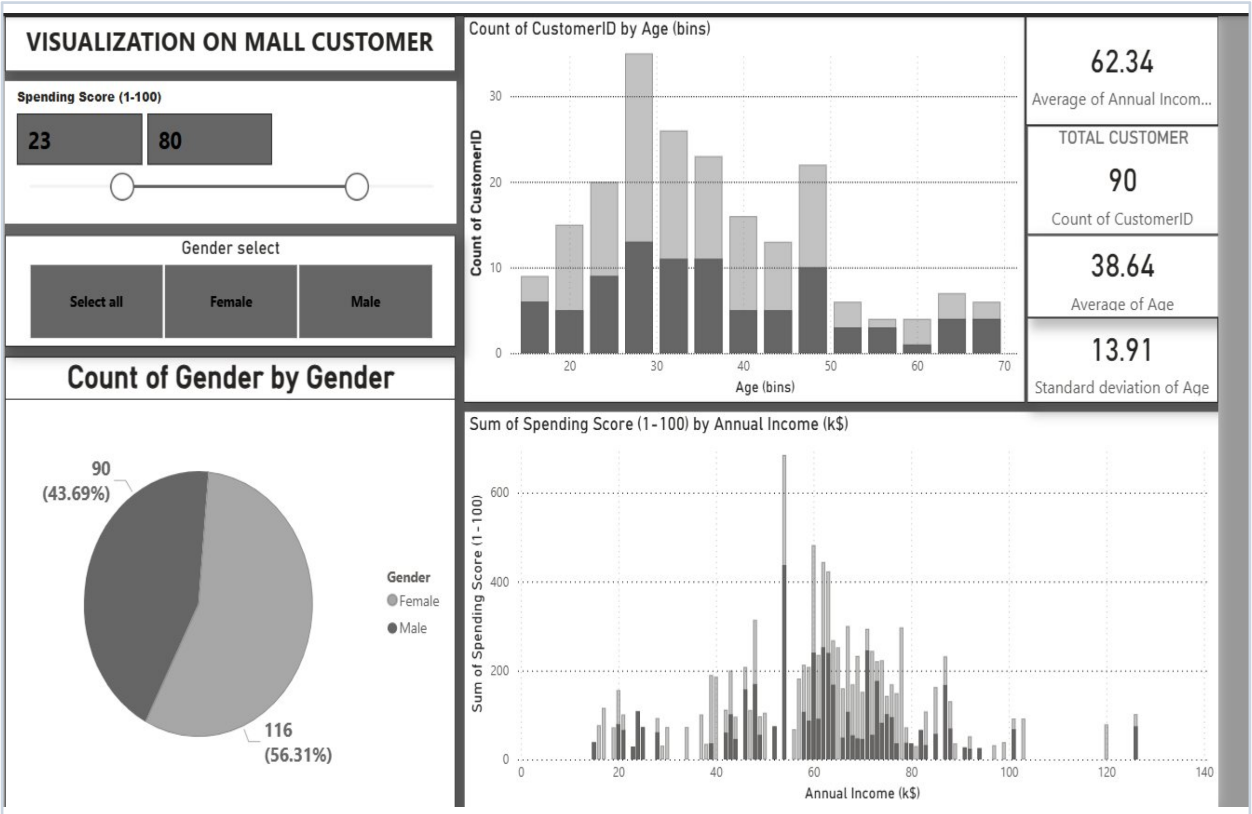

Figure 1: Power BI Dashboard — Mall Customer Segmentation & Spending Behaviour Analysis

Dashboard Component Guide

Component	Type	Key Metric / Insight
Spending Score Slicer	Range Filter	Filters all visuals: default 23–80
Gender Filter	Button Slicer	Male / Female / All
Count of CustomerID by Age	Stacked Bar Chart	Peak cohort: 25–35 years
Count of Gender by Gender	Pie Chart	Female 56.3% (116), Male 43.7% (90)
Average Annual Income	KPI Card	$62,340
Total Customers	KPI Card	200 (filtered: 90 shown)
Average Age	KPI Card	38.64 years
Std Dev of Age	KPI Card	13.91

Component	Type	Key Metric / Insight
Spending Score by Annual Income	Histogram	Spike at $55–65k income band

SECTION 7

Strategic Recommendations

Based on the analytical findings, the following evidence-based strategic initiatives are recommended for the mall management team and its tenant mix strategy.

1. VIP Loyalty Programme for Golden Customers

Introduce an invite-only tier (e.g., 'Platinum Club') for the 32 high-value customers. Offer personalised benefits: dedicated parking, early-sale access, concierge shopping assistance, and birthday vouchers. Aim to increase average visit frequency by 15–20%.

2. Female-First Marketing Strategy

With 56% female customer share and slightly higher average spending scores, campaigns should prioritise female-oriented messaging, influencer partnerships, and female-focused brand acquisition across beauty, fashion, and wellness categories.

3. Mid-Tier Upselling via Targeted Promotions

The 97 mid-range customers (score 36–69) represent the largest conversion opportunity. Deploy bundle offers, cross-category vouchers, and gamified spending challenges to elevate their scores toward the 70+ range.

4. Youth Acquisition Programme (18–24 Cohort)

The 18–24 segment is under-represented (12.5%) relative to their lifetime value potential. A student discount programme, social media activations, and food & entertainment-led footfall drives can grow this cohort by 25–30% over two years.

5. Re-engagement Campaign for Low-Score Customers

Customers with spending scores ≤ 35 and moderate income may be lost to competing retail destinations. A win-back campaign using geo-targeted push notifications, 'We miss you' vouchers, and personalised category offers can recover 20–30% of this segment.

6. Data Infrastructure Enhancement

Extend data collection to capture purchase category, visit recency, and channel preference (in-store vs. online). This will enable RFM (Recency, Frequency, Monetary) modelling and richer predictive segmentation in future analytical cycles.

SECTION 8

Conclusion

This analysis demonstrates that the mall's customer base, while broadly distributed in age and income, exhibits clear and exploitable behavioural clusters. The identification of 32 Golden Customers, the dominance of female shoppers, and the concentration of footfall in the 25–44 age band together provide a strong foundation for data-driven commercial strategy.

The mid-tier spending cohort represents the most scalable opportunity: with the right incentive design, even a modest uplift in their spending scores would materially improve overall mall performance metrics. Meanwhile, the Golden Customer segment demands a bespoke relationship management approach to protect and grow this highest-revenue cohort.

Future analytical work should incorporate temporal data, basket-level transaction records, and machine learning-based clustering (e.g., K-Means) to refine segmentation boundaries and improve the predictive accuracy of customer lifetime value models.

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